

SAATHI / FEATURE GUIDE

What Saathi does today and what comes next

A practical guide to the Android cognitive launcher, the caretaker portal, and the proposed context-recovery roadmap.

STATUS	MEANING
Built	Implemented in the current source tree; some items are uncommitted and need a fresh install or live verification.
Partial	A usable first slice exists, but the complete PDF behavior or end-to-end validation remains.
Planned	Described in the feature plan; not implemented as a working feature.

Review date: 18 September 2026. This document describes code, not a clinical claim or a guarantee that every source change is on the connected phone. The newly added context layer is in the working tree; its live phone-and-portal demo still needs verification.

Product goal: turn ordinary phone actions into gentle memory support. Saathi should tell a person what is known, distinguish inference from fact, and help them resume an intended task.

Basis: the 10-page Saathi Cognitive Launcher New Feature Plan Expanded PDF, the current patient app and caretaker portal source, and recent Git history, reviewed on 18 September 2026. This guide changes no application code.

1. Patient app - what is built

FEATURE	CURRENT EXPERIENCE
Android HOME launcher	Large Flutter home screen can be chosen as the default Android launcher. Kotlin handles HOME role, installed-app discovery, opening apps, dialer and system intents.
Voice-first home	Saathi greets the patient, offers a large Speak button and periodic check-in questions. The screen also provides large app, family, medicine and daily-task controls.
Talk to Saathi	Text and spoken turns use Android speech recognition and text-to-speech. The companion uses Firebase AI Logic for online replies when enabled and available; local rules answer selected questions and commands.
Phone voice commands	Local parser can open installed apps, Home, Phone, Settings, Maps, Calendar and Contacts. A request to call a person asks for confirmation and opens the dialer.
Memory Passport	Editable profile plus family, places, memories, routines, medicine and activities. Entries can include notes and photos; a demo profile is available.
Family recognition	Photo- and relationship-based recall questions use Passport people, with progressive hints and recorded outcomes.
Video recall	Patient can watch a video and answer immediate and delayed recall questions. Recall results are saved.
Memory games	Passport-based family, medicine, routine and life-memory questions offer category choices, mixed play, hints and difficulty options.
FEATURE	CURRENT EXPERIENCE
Medicine support	Medicine schedule from Passport, local reminder notifications, read-aloud details, taken confirmations and medicine recall prompts.
My Day	Timeline from routine and medicine entries; patient can confirm tasks and hear the day read aloud. Completion state is stored locally and can sync when enabled.
SOS and phone	Large help button asks before opening a phone action. It is not an automatic emergency-service call.
Local records	Drift/SQLite stores Passport, cognitive records, video sessions, settings, daily completions and an outgoing sync queue.

2. New context-recovery layer

This is the major addition described by the PDF. It is present in the current working tree, but the complete live demonstration has not yet been certified.

FEATURE	CURRENT SLICE / LIMIT
Action history	With patient consent, the app logs launcher app opens, stated purposes, home returns, Saathi queries and My Day confirmations. Local history is limited to 30 days or 500 events; turning it off clears it.
Intent before app open	For selected launcher app openings, the patient can choose a reason such as watching an exercise video, sending a message or listening to music, or choose Just open it.

FEATURE	CURRENT SLICE / LIMIT
Why am I here?	Explains a recent supported action and includes a 'Why Saathi thinks this' evidence section. If evidence is insufficient, it says so.
Memory Breadcrumbs	Shows recent events as plain-language timeline items rather than raw technical log entries.
Repeated action prompt	Three opens of the same app within ten minutes are recognized. The prompt appears when the patient opens Why am I here?; it is not yet a proactive assist.
Take me back	Can reopen the last supported app or explicitly shared web link. It does not yet restore a specific screen, message or unfinished in-app task.

FEATURE	CURRENT SLICE / LIMIT
Optional UsageStats	After separate Android permission and in-app consent, foreground app timing can provide context. Timing does not reveal content or purpose.
Optional notifications	After separate Notification Access and in-app consent, recent WhatsApp sender/preview signals are kept briefly on the phone. Current reasoning uses sender timing; it does not read chat history.
Share to Saathi	A user can explicitly share a web link. Saathi asks before storing that one URL and can later offer to reopen it.
Daily narrative	Local short summary counts selected actions. The patient can explicitly share the displayed summary to the linked caretaker portal; raw event logs and message previews are not uploaded.

3. Caretaker tools and cloud link

FEATURE	CURRENT EXPERIENCE / LIMIT
Phone caregiver screen	Shows records and care information, provides cloud-sync controls and a phone link code. The daily narrative is available here for patient-approved sharing.
Web portal	Separate dashboard has Overview, Cognitive Records, Medicine & Routines, Memory Passport, Alerts, Daily Notes, Weekly Report and Settings pages.
Passport sync	Linked phone and portal exchange Passport changes through Firestore. Portal editing covers profile, family, routines/medicines and places; phone listener updates local data and reminders.
Care data sync	Opt-in Firestore sync sends Passport, cognitive records and daily completions. An outbox supports retry and a background worker runs when network and battery conditions allow.
Remote checkup	Portal can send a checkup request; the phone receives it and opens Passport-based memory games. The portal can monitor completion when linked.
Daily narrative display	Portal listens for the patient-selected daily summary and shows it on Overview. Sharing must be initiated on the phone.
Portal notes/export	Caretaker notes and JSON export exist. Some notes/settings and several charts, alerts and report examples are browser-local or demo content, not verified live clinical trends.

Cloud behavior depends on Firebase configuration, permissions, network availability and the correct phone-to-portal link. Saathi must never present a dashboard chart or alert as a diagnosis.

4. What still needs building - core PDF roadmap

PRIORITY	REMAINING BUILD	DONE WHEN
1	Person context cards	A reliable Passport match connects a recognized sender to a saved relationship, photo and safe contact action; uncertain matches ask rather than assert.
2	Complete recovery targets	Take Me Back can resume a Saathi screen, saved reminder or other supported target, while clearly saying when an external app cannot be restored.
3	Proactive gentle help	Repeated action detection offers calm context at the moment of need, with dismiss/continue controls and frequency limits.
4	Confidence and provenance	Context engine records source and confidence for derived explanations; the screen clearly separates observed facts, stated intent and tentative inference.
5	Routine drift	A patient-specific baseline identifies meaningful routine changes and offers a gentle prompt. No diagnosis, severity score or automatic clinical conclusion.
6	Full demo verification	On a phone and linked portal: save an intent, open app, return Home, ask Why, reopen target, view breadcrumbs, share a narrative, and see it on portal.

5. Proposed next releases

RELEASE	SCOPE
A - Trustworthy recovery	Finish person cards, recovery targets, confidence labels and the end-to-end demo. Test permission revocation, sparse evidence, offline use and no-fabrication wording.
B - Gentle assistance	Add progressive hints for navigation, confusion episode detection, an I am Lost mode and a short memory replay built from consented events.
C - Caregiver support	Add patient-approved remote cues, stronger narrative controls and clear separation of real synced data from portal demo panels.
D - Personalized context	Evaluate routine drift, routine echo, familiarity links, adaptive home layout and opt-in environment cues only after the event foundation is reliable.

The implementation order follows the PDF's dependency: dependable local event evidence first, then recovery UI, gentle intervention, optional Android signals, caregiver summaries and finally advanced inference.

6. Future concepts from the PDF addendum

Recommended differentiated experience - planned

CONCEPT	INTENDED PATIENT VALUE
Progressive Assistance Mode	Start with a small hint, then highlight a likely next step, then offer a direct action only when needed.
Memory Provenance	Expose the real evidence behind every reconstructed context. A basic evidence sentence exists; structured provenance remains.
Confusion Episode Detection	Recognize a sequence of possibly lost navigation and offer help without labeling or diagnosing the person.
Memory Capsule	Save a short, evidence-based recap after meaningful activity.
Familiarity Graph	Connect enrolled people, places, routines, apps and repeated activities with source-aware relationships.
I am Lost mode	Show a few large recovery actions: family, location, previous task, today and Home.
Digital Familiarity Firewall	Offer a trusted-family check when a sensitive sequence is unusual; requires careful false-positive and privacy testing.
Caregiver Remote Cue	Let a caretaker send a simple suggestion card while the patient keeps control of the phone.
Context Confidence Score	Use confidence internally to choose direct wording versus a clarifying question.
Environment Context	Use only explicitly permitted time, Wi-Fi, Bluetooth or broad place cues to suggest relevant routines.
Routine Echo	Compare today's confirmed sequence with a personal baseline rather than a fixed alarm alone.
Personalized Difficulty	Simplify the parts of the launcher that repeatedly cause difficulty.

Longer-term research or staged demos - planned

CONCEPT	WHAT A SAFE PROTOTYPE WOULD SHOW
Face-to-Memory Recognition	Recognize only consenting, enrolled family photos; privately show saved relationship and memory cues.
Object Memory Assistant	Recognize an object and connect it to the patient's saved, verified routine.
Room-Based Recovery	Use opted-in beacons or predefined demo zones to surface routine cues.
Voice Familiarity	Use an enrolled trusted recording as a memory cue, with explicit recording consent.
Memory Replay	Answer 'What happened today?' from a source-labeled activity timeline.
Conversation Bookmark	Let a patient or caretaker deliberately save a short important note; do not continuously record conversations.
Adaptive Home Screen	Reorder familiar tasks by time, routine and observed difficulty while keeping navigation predictable.
Lost-Task Recovery	Offer Continue, Start Over or Ask Family after a supported confusing sequence.
Trusted Reality Check	Answer medicine and call questions only from confirmations or observed events; otherwise say 'I cannot verify'.
Cognitive Digital Twin	Long-term, privacy-conscious personalization of normal routines and relationships for assistance, never diagnosis.

7. Claims, boundaries and review checklist

RULE	PRACTICAL MEANING
Evidence before explanation	An app open proves an app opened, not what was read or why. Notification timing can support a possibility, not a certain purpose.
Separate permissions	Action memory, Usage Access, Notification Access, speech and cloud sync need clear controls. Optional inputs can be disabled.
Patient control	Daily summary sharing is explicit. Caregiver cues should remain suggestions. Calls and sensitive actions require confirmation.
No diagnosis	Scores, prompts, drift and confusion signals support daily assistance and review; they do not diagnose dementia or deterioration.
Real versus demo	Portal mock charts, alerts and report examples must be labeled before a live presentation. A successful code build does not prove a live Firebase exchange.

Suggested acceptance walkthrough

1. Open Saathi and choose an intent before opening an app. 2. Return Home and ask Why am I here? 3. Confirm the explanation quotes the saved purpose and evidence. 4. Use Take me back. 5. Open the same app three times and assess the wording. 6. Review Memory Breadcrumbs. 7. Approve a daily narrative on the phone and confirm it appears on the linked portal. 8. Turn off optional permissions and confirm the associated context disappears.